

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CLASS TEST - 02

Course Code:	DSA05	Course Title:	Query Processing for Data Science in Real Time Applications
CO Assessed:	CO4	Assessment:	Tool 2 – Class Test - 02
Weightage:		Total Marks:	25 Marks
CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.. (BL4 and BL5)			
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Section / Batch:	Slot-A 2024	Date of Submission:	18 – 06 – 2026
Faculty In-charge:	Dr. K. Veena Devi	Assessment Mode:	Individual Submission

Code of Conduct

I __P.Sunayana(192424303)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: __P.Sunayana__

Q1 (5 Marks)	Q2 (5 Marks)	Q3 (5 Marks)	Q4 (5 Marks)	Q5 (5 Marks)	Total (25 Marks)

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Course code & name:- DSA0501 - Query processing using Data Analysis.

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1) Data cleaning of the Customer Database:-

A suitable sequence of data cleaning steps would be:-

Step 1:- Inspect the dataset:-

First, examine the dataset to identify inconsistent formats, duplicate records, missing values, and other errors.

Why necessary:- It helps understand the quality and extend of data problems.

Step 2:- Standardize data format:-

Converts, all dates from formats such as DD/MM/YYYY and MM-DD-YYYY into one standard format, preferably YYYY-MM-DD.

Why necessary:- A consistent format allow dates to be sorted, filtered, compared, and analyzed correctly.

Step 3:- Identify and remove duplicate customer records.

Compare records using fields such as customer ID, name, email address, or phone number.

Remove extra duplicates and carefully review possible

Duplicate variations.

Why necessary:- Duplicate records can cause the same customer to be counted multiple times.

Step 4:- Handle missing phone numbers:-

Identify records with missing phone numbers. Depending on the purpose of analysis, missing values can be:

1) Filled from another reliable source.

2) Marked as unknown.

3) Left as missing.

4) Removed if the phone number is essential.

Step 5:- Validate the cleaned data:-

Finally, check whether data is consistent, duplicates are removed, and missing values are properly handled.

Validation ensures that the cleaning process has introduced new errors.

2) Analyse of ₹ 15,000,000 as a potential outlier.

Methods to verify the outlier:-

1) Visual inspection:-

Use:- 1) Box plot.

2) Histogram.

3) Scatter plot.

A Box plot can clearly show whether ₹ 15,000,000 lies far away from the majority of observations.

2) Interquartile Range (IQR) method:

Calculate:-

Q_1 = First Quartile.

Q_3 = Third Quartile.

$$IQR = Q_3 - Q_1.$$

Outliers are usually values below:-

$$Q_1 - 1.5 \times IQR$$

(a) above:-

$$Q_3 + 1.5 \times IQR$$

If ₹15,000,000 exceeds the upper limit, it can be significantly classified as an outlier.

3) Z - score:-

Calculate how many standard deviations the value is from the mean.

→ A very large Z - score may indicate an unusual observation.

4) Verify the original data:-

→ A typing error, such as an extra zero.

→ A data - entry error.

→ A currency conversion issue.

→ A genuine large scales transaction.

Impact of keeping a value:-

If the value is genuine, it should generally be retained

→ Increase the mean significantly.

→ Increase the standard deviation.

→ Distort graphs;

→ Create misleading values.

Conclusion:- The value should not be removed automatically. It must first be statistically analyzed and verified using the original business records.

3) Fuzzy Matching (vs) Regular Expression matching.

Given the names:-

→ John Smith.

→ Jan Smith.

→ J. Smith.

→ John Smith.

and email addresses that must follow:-

name@Company.com.

Fuzzy matching:-

Fuzzy matching compares strings based on their similarity:-

- 1) John Smith and Jan Smith → highly similar
- 2) John Smith and J. Smith → possibly similar
- 3) John Smith and John Smith → highly similar

It can use methods such as:-

→ Levenshtein Distance.

→ Jaro-Winkler Similarity.

→ Similarity percentage.

Best use:- Name matching and duplicate detection.

Reason:- Names may contain spelling mistakes, abbreviations, or small variations.

Regular Expression Matching:- (RegEx):-

RegEx is used to check whether text follows a specific pattern.

An example pattern is:-

$^{\wedge}[A-Za-z0-9._+\%]+@Company\.\com\.$

This checks that:-

→ The email contains a valid username input

→ It contains @

→ This domain is exactly company.com

Examples:-

1) John@Company.com → valid

2) JohnCompany.com → Invalid

Comparison:-

Technique	Best use	Reason
Fuzzy matching	Customer name matching	Handles spelling difference and similar names.
RegEx matching	Email validation	Checks whether data follows a fixed pattern.

Conclusion:- Fuzzy matching is more suitable for names,

while RegEx is more suitable for validating email formats.

- 4) Exploring correlation between Advertising Expenditure and product sales:-
The dataset should be organized by variables
- Region.
 - Advertising expenditure.
 - Product sales.
 - Month (or) time period, if available.

Step 1:- Group the data by region:-
Analyze each region separately to identify regional differences.

Step 2:- Calculate Correlation:-

Use a correlation coefficient such as Pearson's

Correlation coefficient to measure the relationship between advertising expenditure and sales.

Recommended visualizations:-

Scatter plot:-

→ Y-axis :- product sales.

→ X-axis :- Advertising expenditure.

Each point represents an observation.

A trend line can also be added.

Correlation Heatmap:-

A heatmap can show correlations between multiple numerical variables such as:-

- Advertising expenditure.
- Sales.
- Profit.

→ Number of Customers.
Interpretation:-

Correlation	Meaning
Strong positive Correlation	Higher advertising expenditure is associated with higher sales.
Weak correlation	Advertising expenditure has little (or) inconsistent relationship.
Negative Correlation	Higher advertising expenditure is associated with lower sales.

Important:- Correlation does not always mean causation. Other factors such as product price, customer demand, season and regional population may also influence sales.

5) Recommended visualizations for City Administration

Data:-

Datatype	Recommended visualizations	Justification.
Monthly traffic volume	Line chart	Shows changes and trends in traffic over time.
Accident locations	Geographic map / Heat map	Identifies accident-prone areas and hotspots.

population distribution across zones	choropleth Bar chart	- on levels across different zones.
public feedback comment	word cloud	Highlights frequently occurring words common public concerns.

1) Monthly Traffic volume - line chart:-
A line chart is suitable because that data is collected over time.

Benefits:-

- Shows increasing / decreasing traffic trends.
- Helps identify peak traffic months.
- Helps in transportation planning.

2) Accident locations:-

A map with accident markers / Heat map can show where accidents occur frequently.

Benefits:-

- Helps identify hotspots.
- Helps authorities prioritize road safety measures.
- Supports better traffic management.

3) Population Distribution:-

- A choropleth map displays population levels across geographical zones.
- A bar chart can also compare the population of each zone.

Benefits:-

- shows highly populated and low - population zones.
- Supports infrastructure and resource planning.

4) Public Feedback Comments:-

A word cloud display frequently used words with greater emphasis (as more common terms).

for example :-

Traffic, Road, Garbage, Water, Safety.

may indicate important public concerns.

Benefits:-

- Provides a quick summary of large amounts of text.
- Helps decision-makers identify common issues.